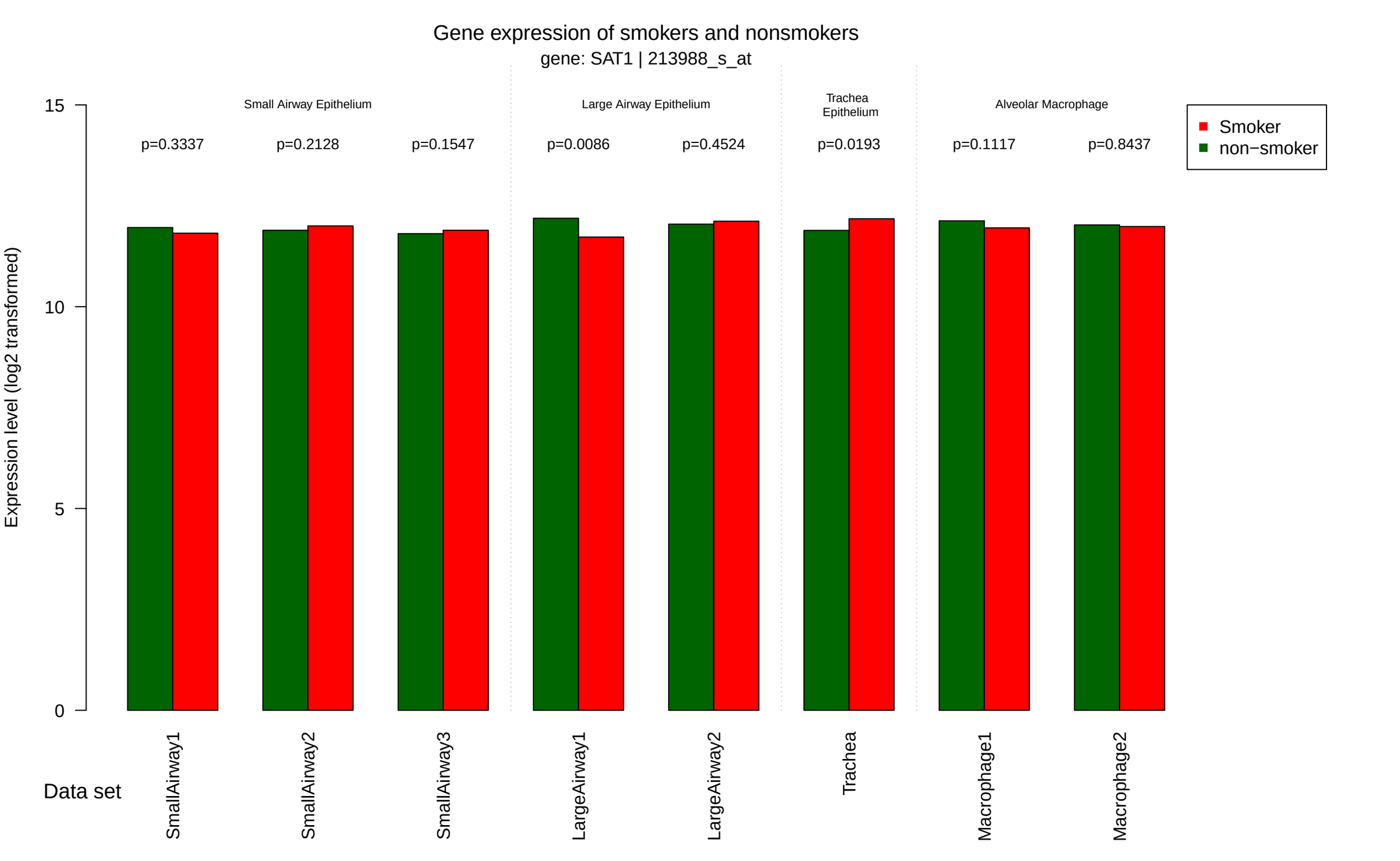
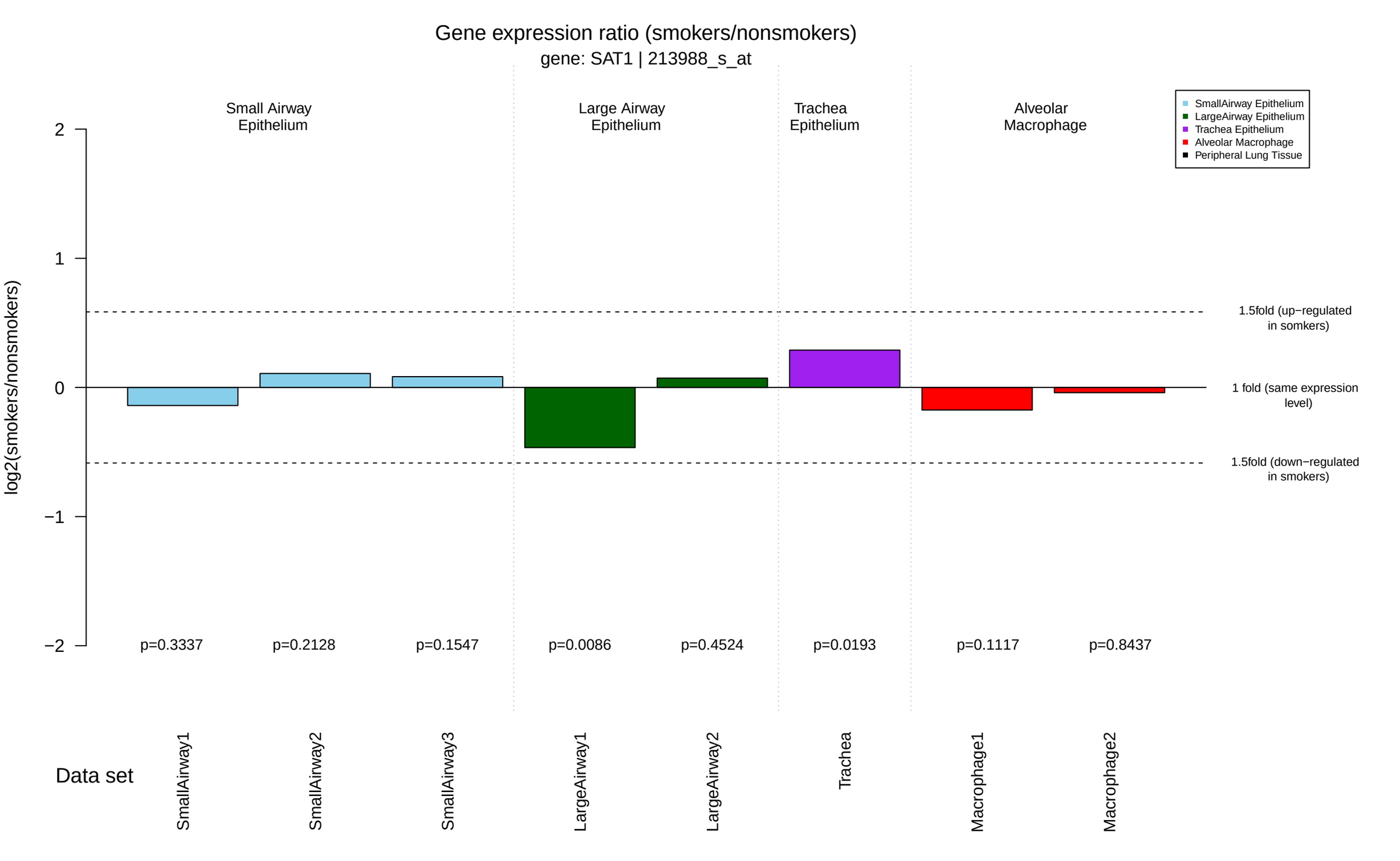
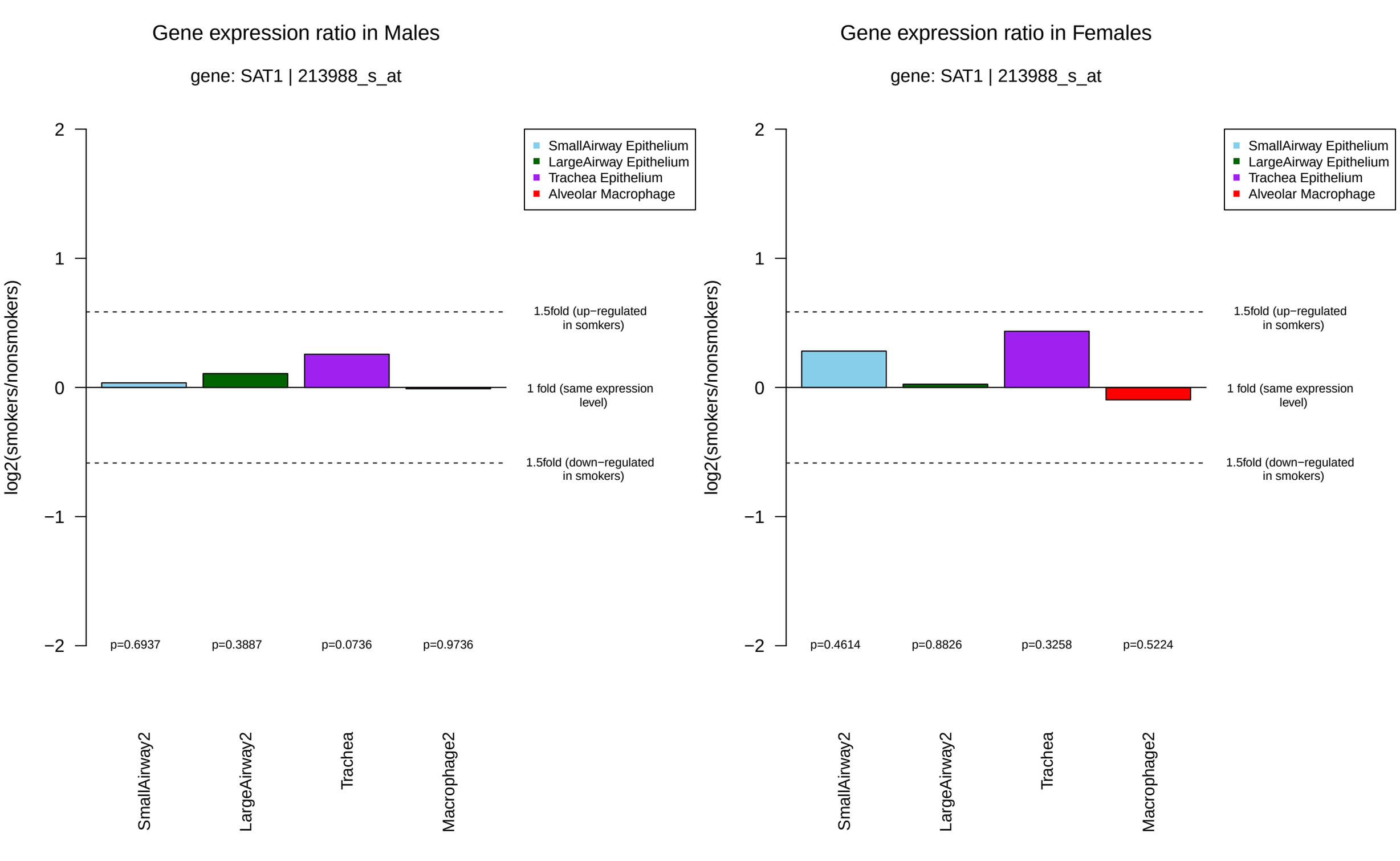
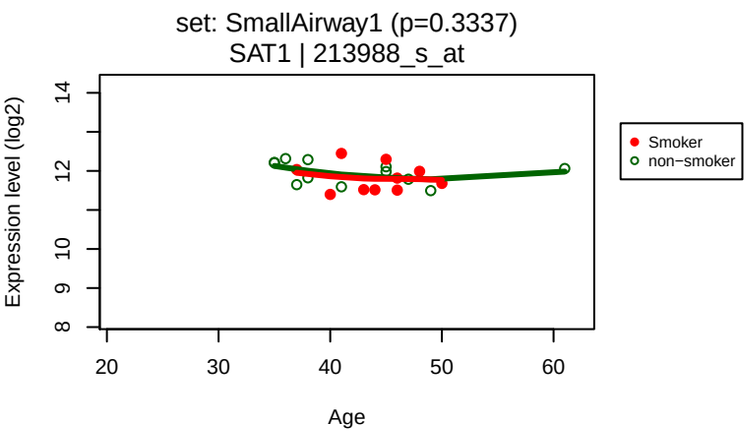
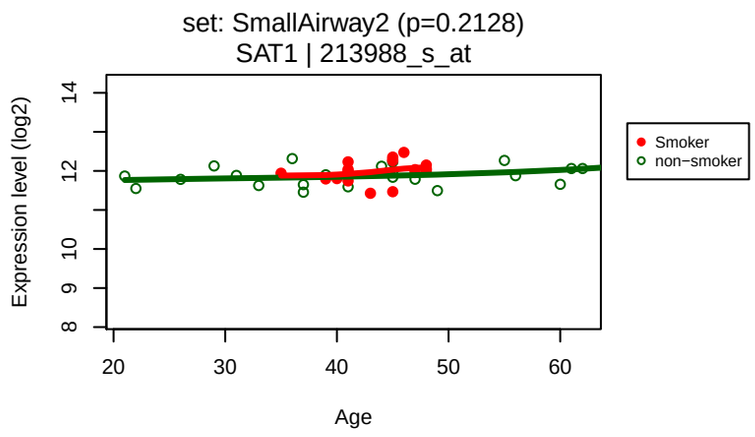
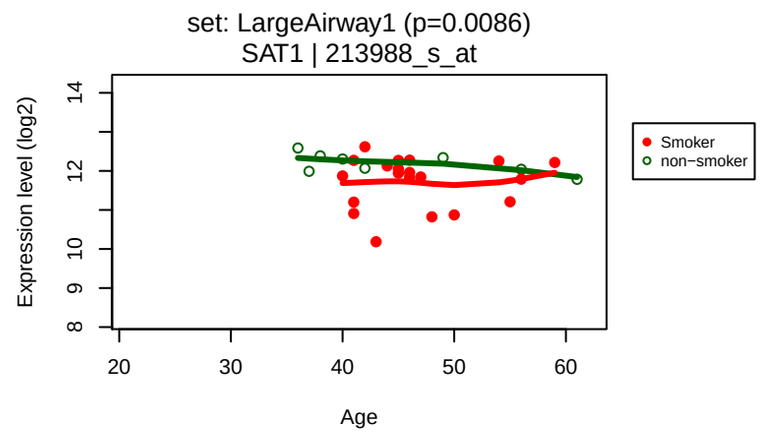

Gene expression of smokers and nonsmokers
gene: SAT1 | 213988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3337
p=0.2128
p=0.1547
p=0.0086
p=0.4524
p=0.0193
p=0.1117
p=0.8437
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

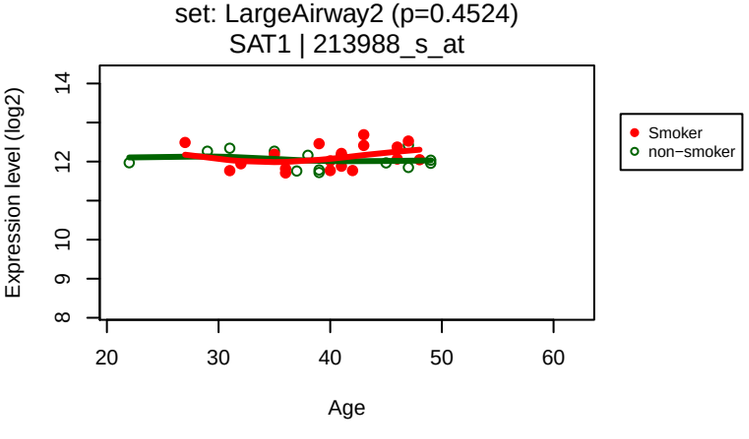

Gene expression ratio (smokers/nonsmokers)
gene: SAT1 | 213988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3337
p=0.2128
p=0.1547
p=0.0086
p=0.4524
p=0.0193
p=0.1117
p=0.8437
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

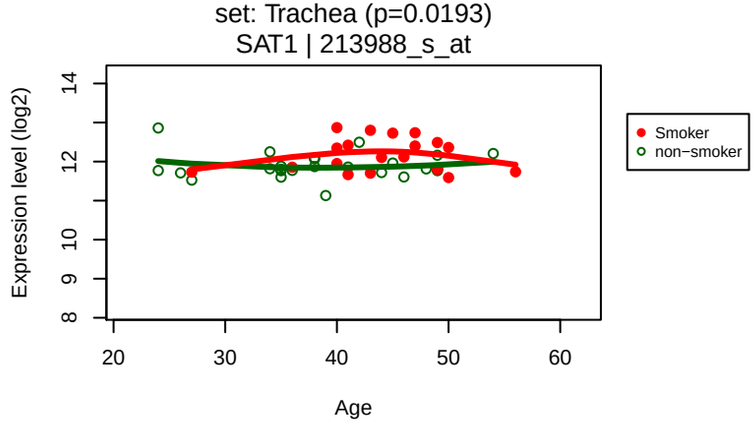

Gene expression ratio in Males
gene: SAT1 | 213988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6937
p=0.3887
p=0.0736
p=0.9736
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SAT1 | 213988_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4614
p=0.8826
p=0.3258
p=0.5224
SmallAirway2
LargeAirway2
Trachea
Macrophage2

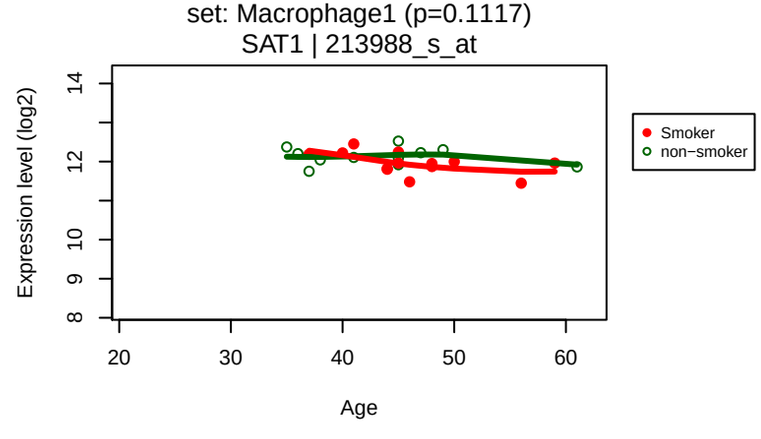

set: SmallAirway1 (p=0.3337)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker

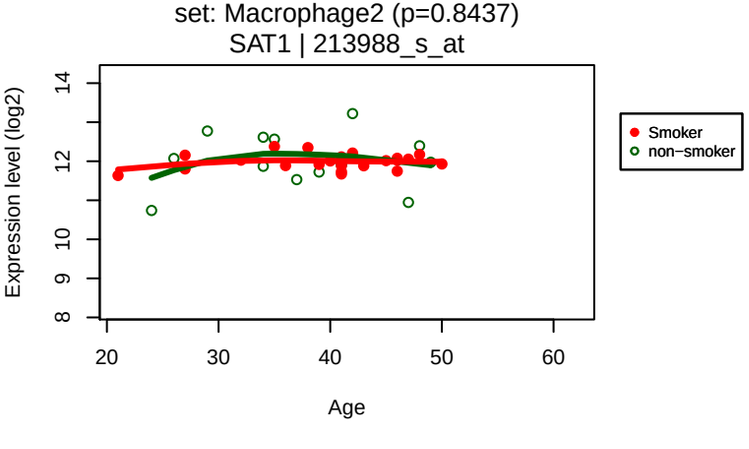

set: SmallAirway2 (p=0.2128)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker

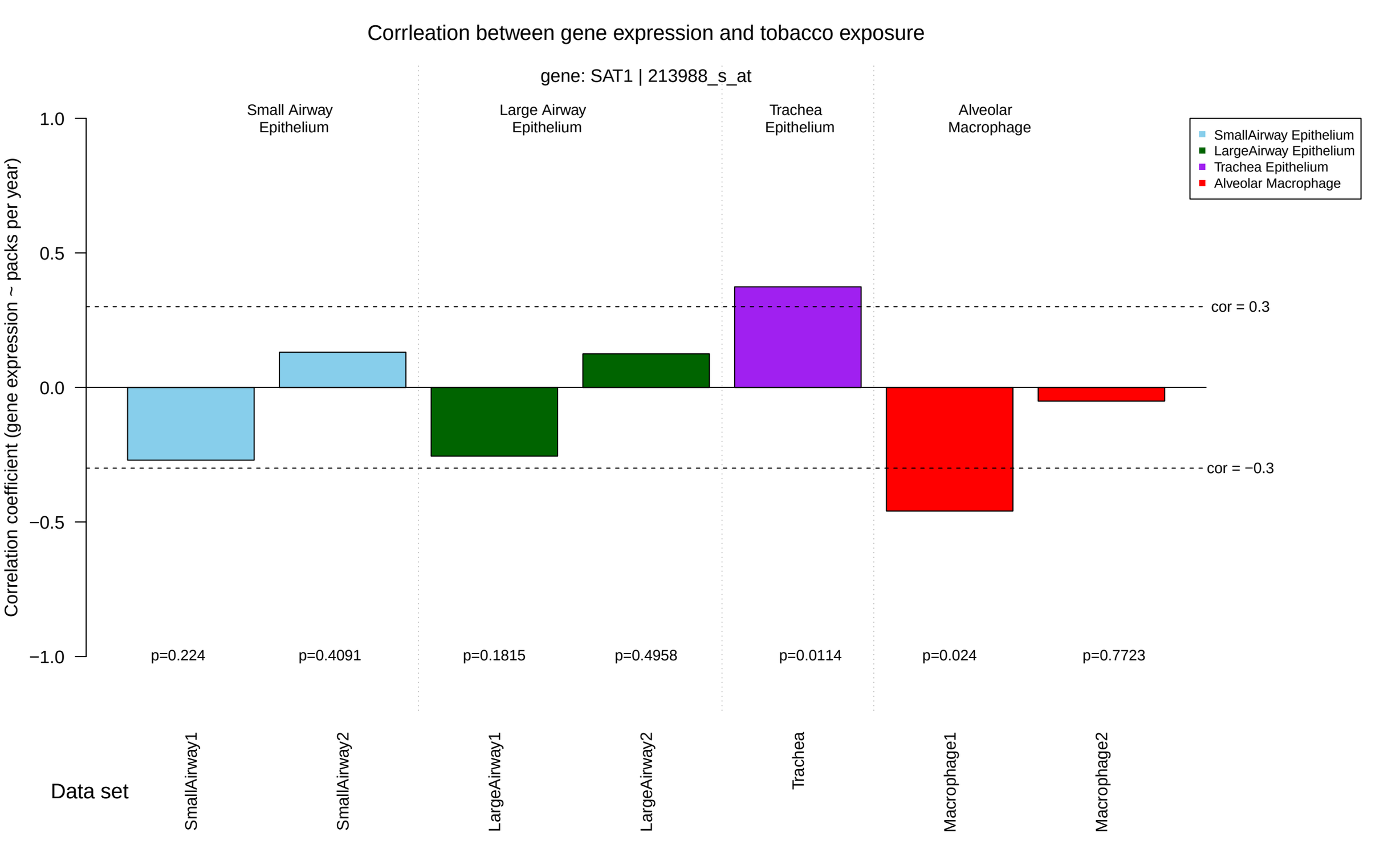

set: LargeAirway1 (p=0.0086)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4524)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0193)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1117)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8437)
SAT1 | 213988_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SAT1 | 213988_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.224
p=0.4091
p=0.1815
p=0.4958
p=0.0114
p=0.024
p=0.7723
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2